KCE PATHWAYS · GRADES 4-5 · UNIT 3: BUILDING THINGS · WEEK 13 · SCIENCE

Forces and Motion

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

What Do I Already Know?

Answer with what you know right now. It is OK to guess!

1. What makes a moving object slow down and stop? Give an example.

2. Predict: Will a toy car roll farther on carpet or on a smooth floor? Why do you think so?

3. Draw a tool or machine that helps people lift or move heavy things. Label how it helps.

4. What is one question you have about forces or motion?

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POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Show What I Learned

Use evidence from today's lesson to explain your thinking.

1. Draw force arrows for a tug-of-war where one team is winning. Label the forces and explain why the rope moves.

2. Use the friction data table: rank the four surfaces from most friction to least, and give two numbers as evidence.

3. Explain the trade-off a ramp or lever makes. Use the words force and distance.

4. Elijah McCoy and Al-Jazari both built machines. How did each one's work use or fight friction or other forces?

★ **Challenge:** Design a fair test: does a heavier toy car roll farther than a lighter one? Name what you would change, what you would measure, and what you would keep the same.

Looking Ahead: Next week you will explore the human body and how its systems work together. Question to bring: your muscles pull on your bones to move you. Which simple machine does your arm act like?